

Appearance Stability Framework (ASF)

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1. Introduction

The Appearance Stability Framework (ASF) is a diagnostic epistemic framework designed to detect latent instability in complex systems. ASF does not aim to predict future system trajectories or optimize control actions. Instead, it evaluates whether a system's apparent stability is structurally supported by its internal coherence.

2. Core Concept

Many complex systems exhibit surface-level stability while internally accumulating divergence, misalignment, or structural stress. ASF formalizes this distinction by separating a system's *appearance state* from its *coherent internal state*. A system is considered appearance-stable if its observed outputs suggest equilibrium, regardless of whether such equilibrium is internally sustainable.

3. Formal Definition

Definition (Appearance Stability): A system is said to be appearance-stable over an interval if its observable outputs remain within bounded nominal thresholds, while its internal state evolution may diverge from any trajectory that preserves long-term coherence or controllability.

Definition (Appearance Stability Divergence): Appearance Stability Divergence occurs when the discrepancy between a system's observable stability and its internal coherence exceeds a critical threshold, indicating latent instability despite continued nominal performance.

4. Epistemic Scope

ASF is epistemic rather than predictive. It is concerned with what is structurally true about a system, not with forecasting specific outcomes. The framework is agnostic to system domain and may be applied to technical, institutional, social, or cognitive systems, provided that observable behavior and internal coherence can be meaningfully distinguished.

5. Non-Claims

ASF does not claim to provide control strategies, policy prescriptions, or optimization mechanisms. It does not assert inevitability of failure, nor does it quantify time-to-collapse. Its sole function is diagnostic: to reveal whether apparent stability is structurally justified.

6. Status

This document represents an initial formalization of the Appearance Stability Framework. The framework is under active research and refinement.